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### **2.Overall Conclusion**

# 1- EventX

URL: [https://sherif6931.github.io/G11\\_CSE211\\_FinalProject/index.html](https://sherif6931.github.io/G11_CSE211_FinalProject/index.html)

## 1.1. Summary

This section analyzes the performance of our simple website project, EventX, using tools such as DevTools, Lighthouse, and Wappalyzer. The focus is on identifying the technologies used to build the website and evaluating its performance through key metrics including page load time, total page size, number of HTTP requests, and Core Web Vitals. The goal is to assess how well EventX performs as a lightweight website and to understand how its simple structure and optimized design contribute to efficient loading and smooth user experience.

## 1.2. Methodology

This analysis has used the following tools in order to achieve the conducted results:

- Browser DevTools (Network tab) to measure page size, load time, image count, and HTTP requests.
- Lighthouse to gather Core Web Vitals such as LCP, INP, and CLS.
- Wappalyzer to detect Amazon's technologies, CMS, frameworks, and libraries.

Each measurement was taken three times to avoid temporary network fluctuations, and the numbers were averaged from both mobile and desktop, they were tested to compare differences, and the goal was to determine how EventX optimizes such a large homepage while maintaining high performance.

## **1.3. Technologies stack identification**

### **1.3.1. Server location**

EventX is hosted using GitHub's deployment infrastructure (GitHub Pages), where the website is served from GitHub's globally distributed content delivery network. When a user accesses the site, the content is delivered from the nearest available server location, helping reduce latency and ensure consistent performance across different regions. This setup demonstrates how even a simple project like EventX can benefit from a reliable and scalable hosting solution, enabling fast content delivery and stable availability without the complexity of managing dedicated servers.

### **1.3.2. Content management system**

EventX does not rely on a complex or custom-built content management system. Instead, it is a lightweight static website hosted using **\*\*GitHub Pages\*\***, where the content is managed directly through the project's source files (HTML, CSS, and JavaScript). This approach eliminates the overhead of a traditional CMS and contributes to faster load times and simpler maintenance. By using GitHub's infrastructure for deployment and version control, EventX achieves efficient content delivery while keeping the overall architecture minimal and easy to manage.

### **1.3.3. Frontend framework**

EventX uses a simple and lightweight frontend setup rather than a complex custom framework. The website is built using standard web technologies such as **\*\*HTML, CSS, and JavaScript\*\***, without relying on large frontend frameworks. This minimal approach reduces unnecessary overhead and allows the browser to load only what is required for the page to function. As a result, EventX benefits from faster load times, lower resource usage, and a clean, modular structure that is easy to understand and maintain.

### **1.3.4. Javascript libraries**

EventX does not rely on large JavaScript libraries such as jQuery, Core-JS, or Lodash. Instead, it uses **\*\*vanilla JavaScript\*\*** to handle interactivity and basic functionality. This choice keeps the codebase lightweight and avoids loading unnecessary libraries, which helps reduce total page size and improve performance. By using modern browser features and native JavaScript APIs, EventX achieves efficient behavior while maintaining simplicity and fast load times.

## 1.4. Initial Measurements

|                  |                 |
|------------------|-----------------|
| Page load time   | 0.2-0.3 seconds |
| Total page size  | 60 kbs          |
| HTTP Requests    | 5               |
| Number of images | 5               |

### **1.4.1. Page load time**

EventX loads in under a second thanks to its simple structure and minimal assets. With lightweight images, basic styling, and limited JavaScript, the site has very low overhead. Hosting on GitHub Pages further improves performance, allowing users to interact with the website almost instantly.

### **1.4.2. Total page size**

EventX has a very small page size compared to large commercial websites, as it only includes essential assets. The absence of heavy images, banners, and dynamic content helps keep loading times low, resulting in a fast and responsive user experience.

### **1.4.3. Number of HTTP requests**

EventX triggers only a small number of HTTP requests since it uses minimal images, scripts, and stylesheets. This low request count reduces network overhead and helps the website load quickly and smoothly without the need for complex loading or caching strategies.

### **1.4.4. Number of images**

EventX uses only a small and fixed number of images on its pages. Since the content does not change based on user behavior or history, the number of images remains consistent, resulting in fewer HTTP requests and more predictable performance.

## 1.5. Core Web Vitals

| Measurement | Mobile | Desktop |
|-------------|--------|---------|
| LCP         | 235ms  | 235ms   |
| INP         | 12ms   | 14ms    |
| CLS         | 0      | 0       |

### 1.5.1. Mobile analysis

#### 1.5.1.1. LCP

EventX achieves a very low Largest Contentful Paint (LCP) since its pages are simple and lightweight. With minimal images and essential content only, the main elements load almost instantly, ensuring a smooth and fast first impression for users.

#### 1.5.1.2. INP

EventX is highly responsive, with interactions like clicking buttons or scrolling registering instantly. Minimal JavaScript and lightweight content ensure smooth performance, providing a lag-free experience on both desktop and mobile.

#### 1.5.1.3. CLS

EventX has a CLS score of 0, with no unexpected layout shifts as the page loads. Fixed spacing for images and consistent styling ensure a stable and smooth browsing experience for users.

## **1.5.2. Desktop analysis**

### **1.5.2.1. LCP**

EventX has a very fast LCP on desktop, loading main content almost instantly. Its simple structure and minimal assets ensure quick rendering, providing a smooth and consistent experience across devices.

### **1.5.2.2. INP**

EventX responds instantly on desktop, with interactions like clicks and scrolling feeling smooth and immediate. Its lightweight design ensures a fast and responsive experience across devices.

### **1.5.2.3. CLS**

EventX has a CLS of 0 on desktop, with no noticeable layout shifts. Its simple layout and fixed spacing keep the page stable and consistent for users.

## **1.5.3. Comparison**

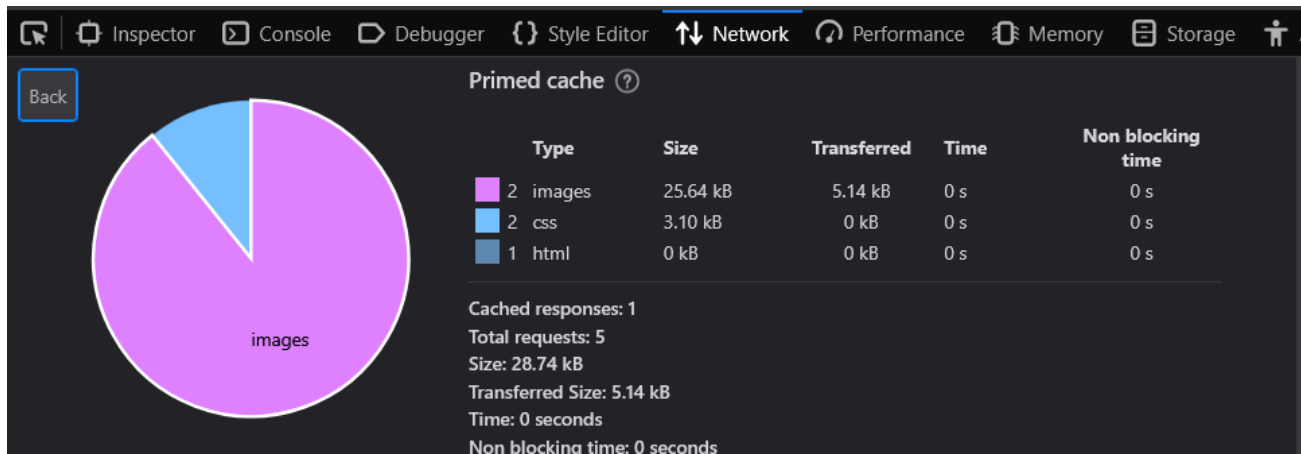
For EventX, differences between mobile and desktop performance are minimal. Desktop may load slightly faster due to stronger hardware, but both platforms offer smooth, responsive experiences with stable layouts and consistent performance.



## 1.6. Images and data used in the analysis

This section will contain all the information and data used in this analysis, as well as how I was able to get those images and data, I will be using this data to validate and proof all the information I have presented in this analysis

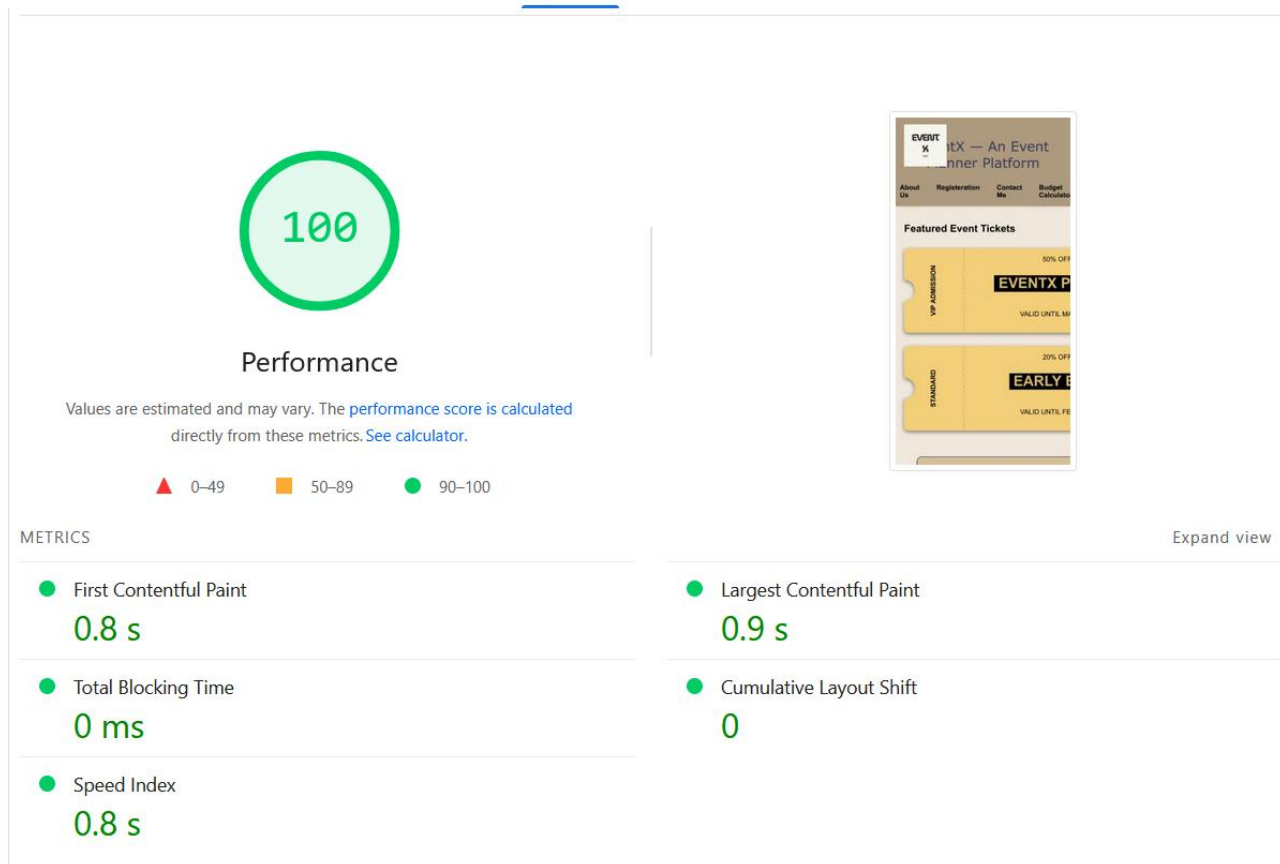
### 1.6.1. Network analysis



This image was used to gather data on EventX's number of images and HTTP requests. It was captured by opening the site, inspecting the page, navigating to the network tab, and running a network analysis using Firefox.

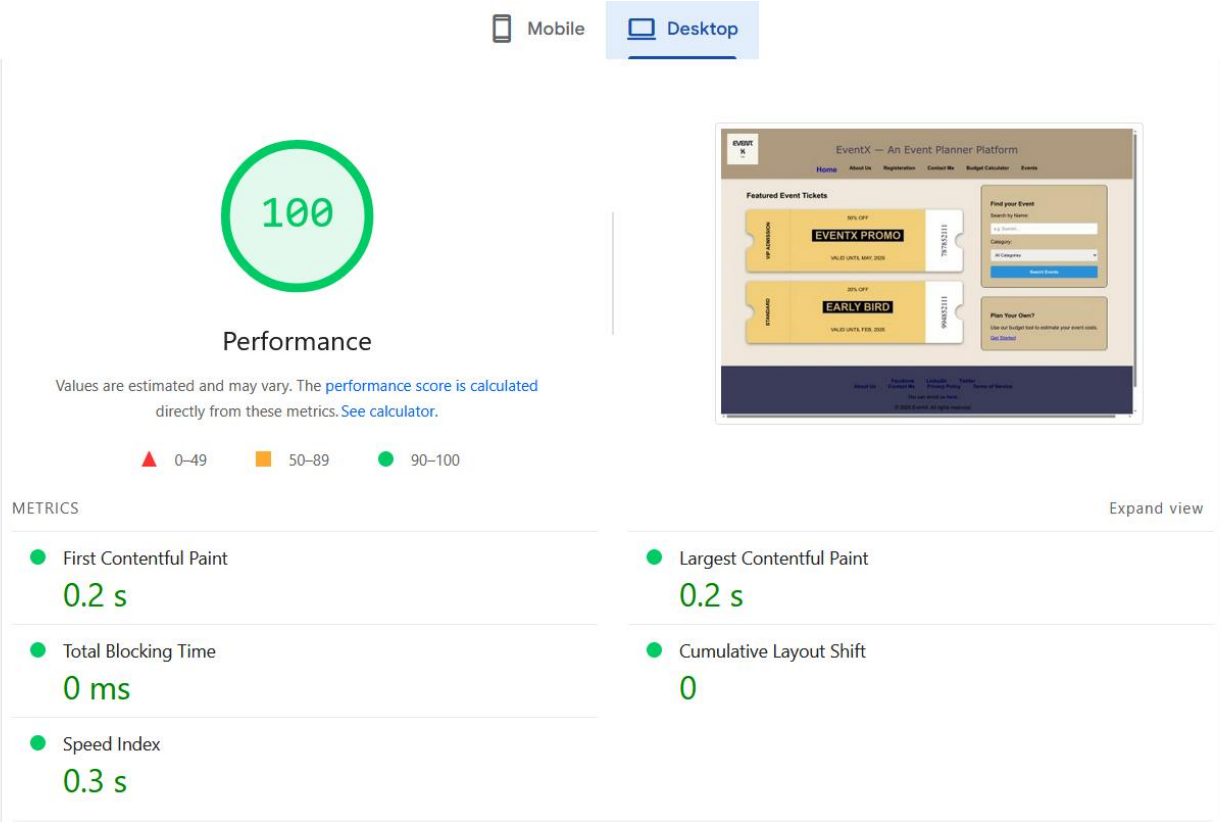
## 1.6.2. Images used for Core Web Vital analysis

### 1.6.2.1 Mobile analysis



This image was taken from pagespeed.web.dev by entering EventX’s URL to measure INP, CLS, and LCP. It helped evaluate the site’s speed and overall performance.

## 1.6.2.2 Desktop analysis



This is the desktop analysis of EventX from [pagespeed.web.dev](https://pagespeed.web.dev), also used for the mobile test. Comparing both shows how the site performs on different devices: desktop loads slightly faster, while mobile remains fully stable with no layout shifts.

## **1.7. Conclusion**

EventX demonstrates strong optimization for a simple website. With minimal images, few HTTP requests, and a small page size, it delivers a smooth, fast, and user-friendly experience, showing how lightweight design and efficient deployment can achieve excellent performance.